

✓ MSDS 570 Visualization and Unstructured Data Analysis

Homework Assignment 1: Introduction to Different Type of Plots

Due Feb 26th, 2023 (11.59 pm CST)

Total 100 points

Homework Submission instructions

To submit this homework:

- Please create a new notebook named as **Name_MSDS570_HW#1**.
- Please write your solutions and submit the notebook files on Blackboard or email me if facing issues in submitting the notebook file.
- Assignments **MUST** be submitted on time.
- This programming assignment **should be done individually**. You can discuss the problems but the codes should not be same.
- Late submission policy will be applied for the late submissions
- Feedbacks will be provided in your solution notebook below the questions

✓ Problem 1 (25 Points)

You are given below diagram that gives information about the road accidents that have occurred over the past two decades during the months of January, April, July, and October:

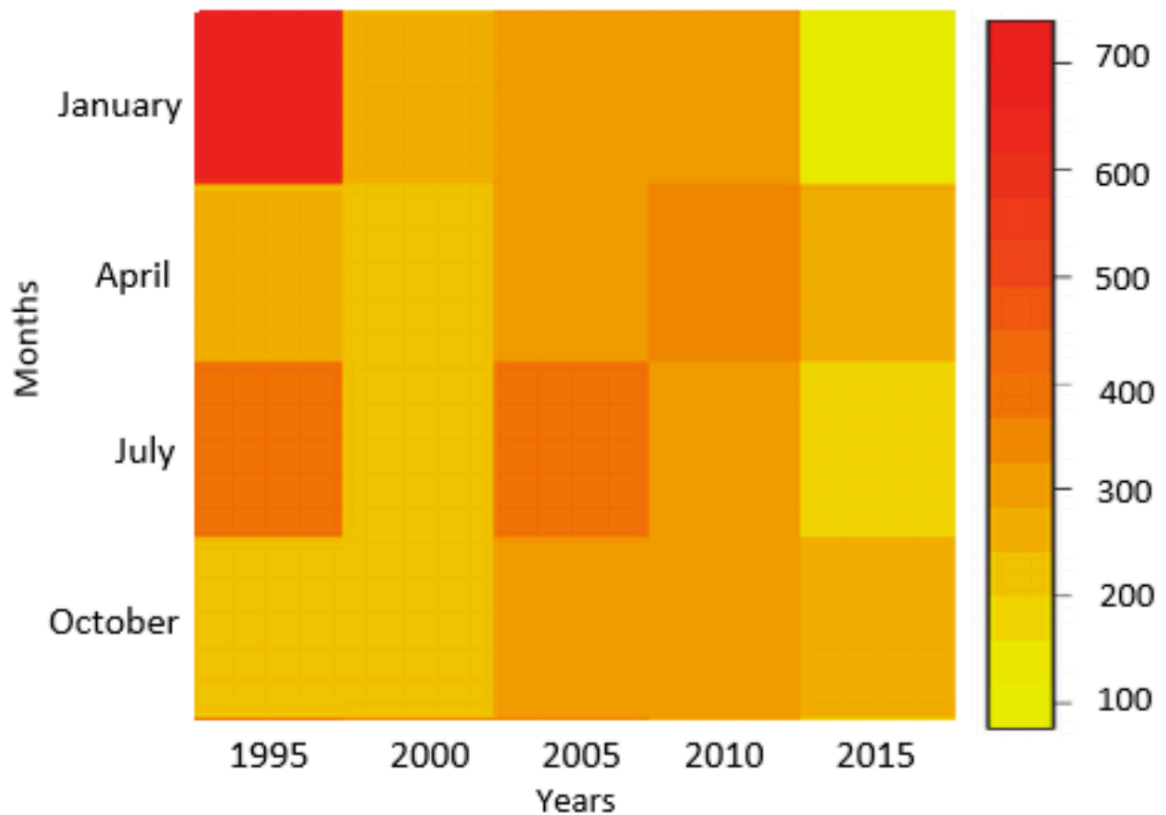


image.png

- (a)** Identify the year during which the number of road accidents occurred were the least.
- (b)** For the past two decades, identify the month for which accidents show a marked decrease:

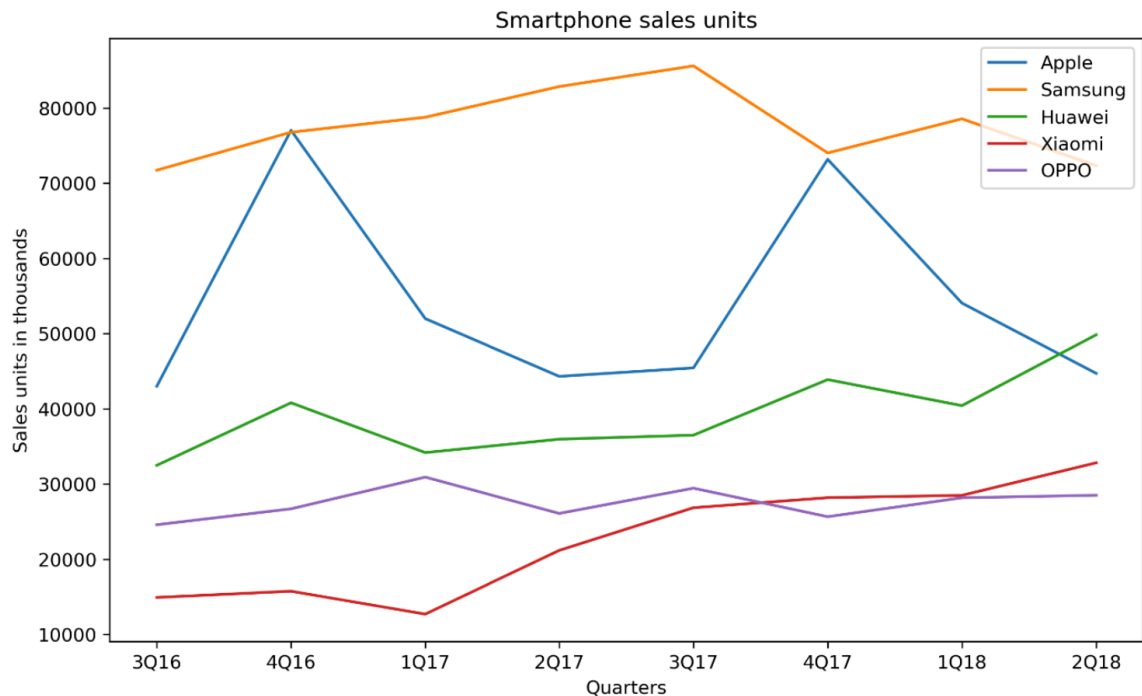
Response

a) The year during which the number of road accidents occurred the least was 2015

b) The month for which accidents showed a marked decrease across the past two decades was JANUARY

✓ Problem 2 (25 Points)

You want to compare smartphone sales units for the five biggest smartphone manufacturers over time and see whether there is any trend:



(a) Looking at the above line chart, analyze the sales of each manufacturer and identify the one whose performance is exceptional in the fourth quarter as compared to the third quarter.

(b) Analyze the performance of all manufacturers, and make a prediction about two companies whose sales unit will show a downward trend and an upward trend.

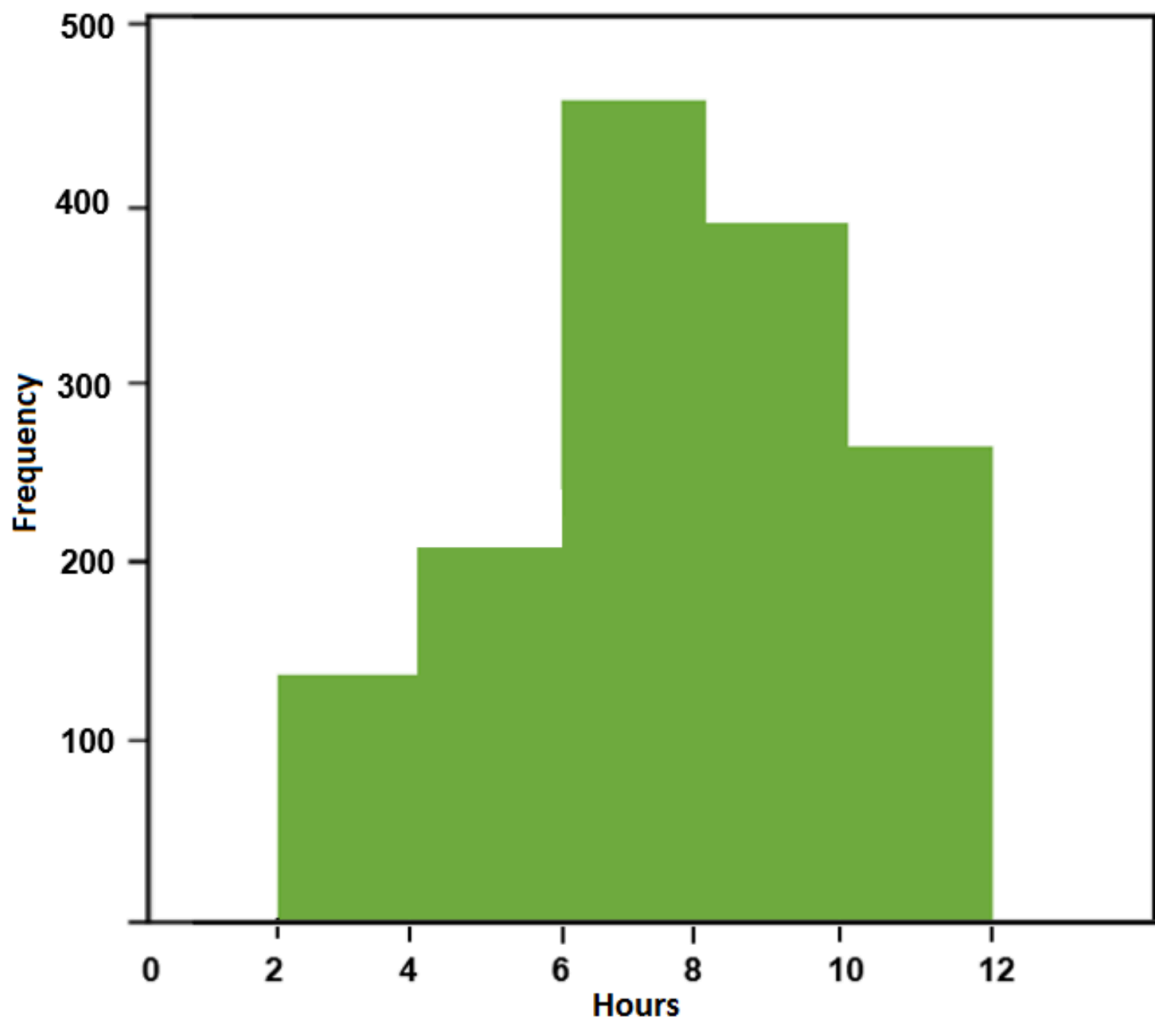
Response

a) Analyzing all 5 smartphone companies. The company with the overall exceptional performance in the fourth quarter as compared to the third quarter is APPLE, there is a significant peak in the line chart graph.

b The prediction of two smartphone companies with a downward and upward trend based on the line chart graph are APPLE which shows a upward trend from the thrid to the fourth quarteer followed by a significant downward trend in the first and second quarter. HUWAWEI is the second smartphone with the biggest upward and downward trend which mirrors APPLE trend based on quarter sales

✓ Problem 3 (25 Points)

You are provided with a histogram that states the total number of trains arriving at different time intervals:



- (a)** By looking at the above graph, can you identify the interval during which the most number of trains arrive?
- (b)** How would the histogram change if the number of trains arriving between 4 and 6 pm were to be increased by 50?

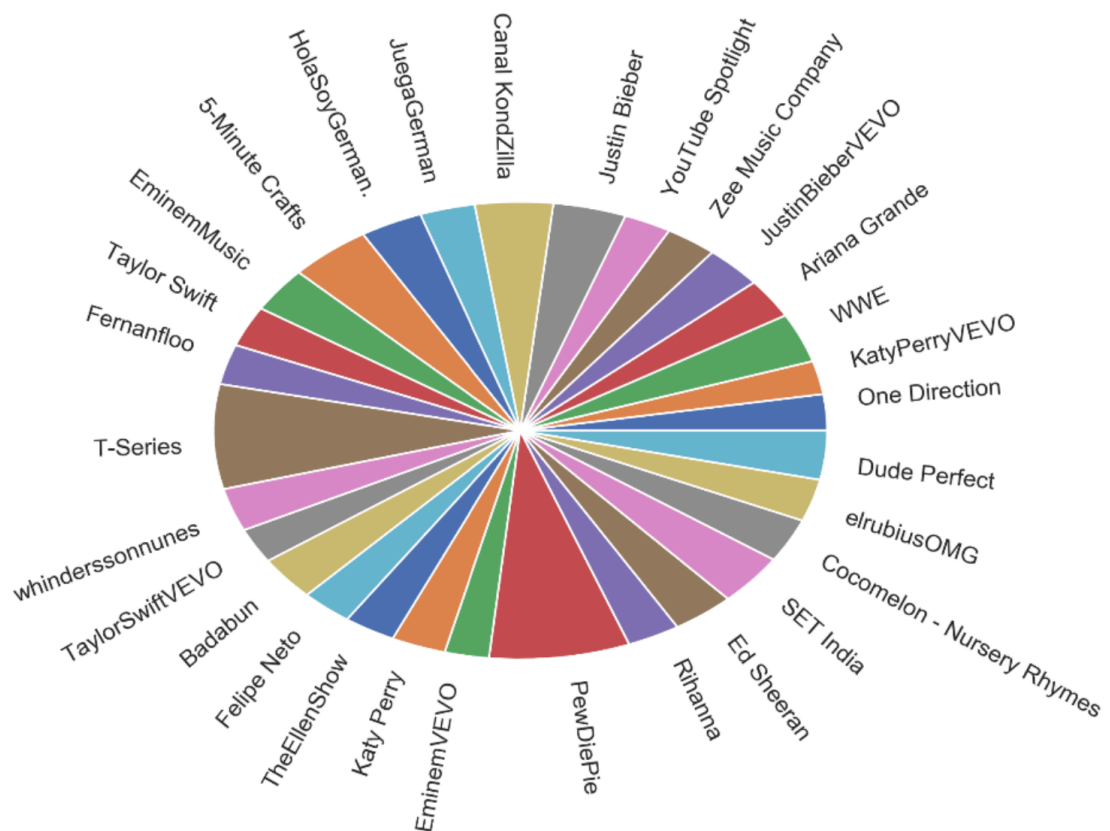
Response

- The interval during which the most number of trains arrive is between 6-8 hours which have a frequency of around 450
- The histogram will change by altering the frequency of train arrival at that time interval. Overall, the histogram will appear to have a more normal distribution (symmetric bell shaped curve)

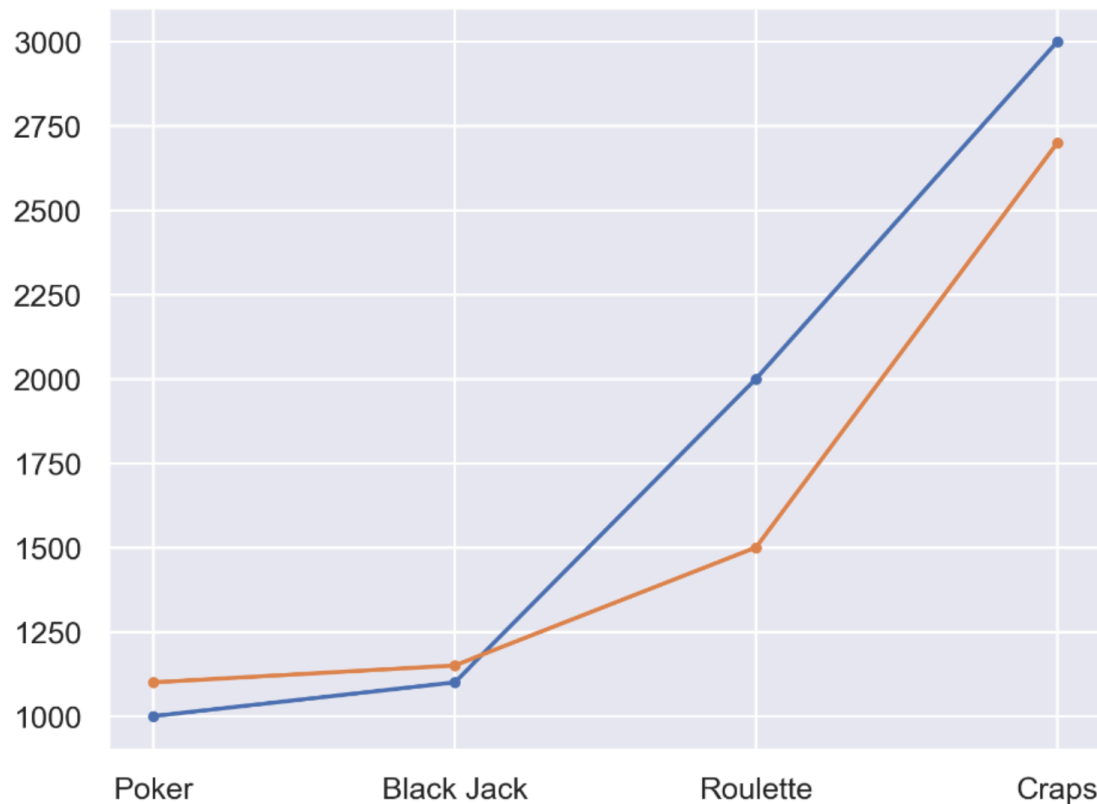
✓ Problem 4 (25 Points)

The following visualizations are not ideal as they do not represent data well. Answer the following questions for each visualization:

The first visualization is supposed to illustrate the top 30 YouTubers according to the number of subscribers:



The second visualization is supposed to illustrate the number of people playing a certain game in a casino over two days:



(a) What are the bad aspects of these visualizations?

(b) How could we improve the visualizations? Propose the right visualization for both scenarios.

Response

a i) The pie chart is too condensed, it lacks the percentage of top subscribers on youtube, color palette is repetitive for some youtubers, the pie chart also lack a label, some dataset are redundant such as "Taylor Swift and Taylor Swift VEVO, ideally they should be clustered as they represent the same artist or youtubers. Since the visualization does not represent the whole subscribers on youtube but rather the top 30. A pie chart is not the most suitable".

a ii) It is recommended to add a vertical y-axis label showing the number of people playing game in the casino, a legend is also suitable and a descriptive title is necessary.

b i) My proposed visualization first aim to clean the data by merging similar youtubers or artist and then finding the appropriate visualization technique that represent the top 30 subscribers on youtube in which I need to take into account the total number of subscribers to create a composition plot such as a pie chart or a donut chart with an appropriate explicit title noting this account for just the "top 30 most subscribed youtubers" rather than the whole youtube subscribers. I will also incorporate a percentage/percentage and ensure the palette are different by attributing different shade of colors. This will also add new data point and show existing youtubers who made no be accounted for due to the redundancy of the first dataset.

b ii) My proposed visualization aim to elucidate not the trend of each game played in the casino but the number of people playing each particular game, which game was popular among each individual. To achieve this I will use a comparison plot. My suitable plot of choice is most likely a bar chart which will show the number of people on different categorical variables (type of game played at the casino) which will include a legend, a title, a x and y axis label

